

The normal distribution

Laboratory 3, DEDP

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1 Objective

Generate normal samples for different μ and σ . Compare their histograms and sample statistics with theory. Compute normal probabilities using `erf()` and estimate them from samples.

2 Theoretical aspects

2.1 The normal distribution

Definition and parameters

The normal distribution describes a random variable that is more likely to take values near its mean μ than far from it, according to a bell-shaped PDF (“Gaussian hat”).

A normal variable is written as:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma > 0.$$

Its probability density function is

$$w_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

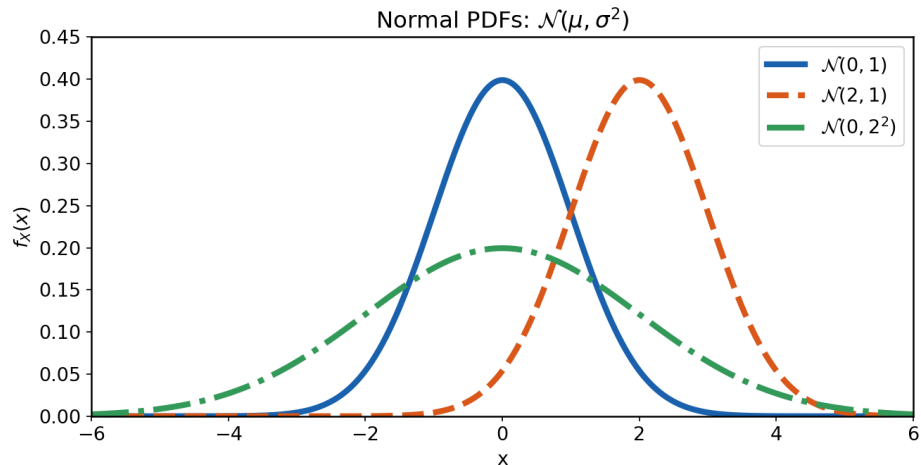


Figure 1: Different normal distributions

The normal distribution has two parameters:

- μ , the **mean** (the “center”), which controls where the peak of the PDF is located;
- σ , the **standard deviation**, which controls the width of the lobe.
- Sometimes we provide the value of σ^2 , the **variance**, instead of σ .

Changing μ shifts the PDF. Larger σ means the function is wider and flatter, smaller σ means narrower and taller.

The **standard normal distribution** is the normal distribution with $\mu = 0$ and $\sigma = 1$:

$$Z \sim \mathcal{N}(\mu = 0, \sigma^2 = 1).$$

Generating normal samples in Matlab

`randn()` generates values from the standard normal distribution $\mathcal{N}(\mu = 0, \sigma^2 = 1)$:

```
z = randn()           % one value from N(0,1)
z = randn(1, 10)      % 10 independent values from N(0,1)
```

For arbitrary μ and σ , scale with σ and shift with μ :

$$B = \mu + \sigma A, \quad \text{where } A \sim \mathcal{N}(\mu = 0, \sigma^2 = 1).$$

```
mu = 5;
sigma = 2;
N = 1000;

x_normal = mu + sigma*randn(1, N);
```

Note: multiply by the standard deviation σ , not the variance σ^2 .

Computing probabilities from the normal distribution

The normal CDF cannot be expressed using elementary functions. Use Matlab's `erf()` function to evaluate it.

For a normal variable $A \sim \mathcal{N}(\mu, \sigma^2)$, the probability that A is in the interval $[a, b]$ is:

$$P(a \leq A \leq b) = F_A(b) - F_A(a),$$

where $F_A(x)$ is the cumulative distribution function (CDF) of A and is computed using the error function `erf()`:

$$F_A(x) = P(A \leq x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma\sqrt{2}} \right) \right].$$

In Matlab:

Example: $A \sim \mathcal{N}(\mu = 10, \sigma^2 = 4)$ and $P(8 \leq A \leq 12)$:

```
mu = 10;
sigma = 2;
a = 8;
b = 12;

F_a = 0.5*(1 + erf((a-mu)/(sigma*sqrt(2))));
F_b = 0.5*(1 + erf((b-mu)/(sigma*sqrt(2))));
p_normal_theory = F_b - F_a
```

The result is approximately 0.6827. We can also estimate it from samples using the method introduced in Lab 2:

```
N = 10000;
x_normal = mu + sigma*randn(1, N);
p_normal_estimated = sum(x_normal >= a & x_normal <= b)/N;
```

2.2 Computing statistics from data samples

Given a vector of samples, we can:

- use `mean()`, `std()`, and `var()` to compute statistics from the data. They estimate the theoretical values μ , σ , and σ^2 , but the precision depends on the number of samples used.
- use the normalized histogram introduced in Lab 2 to compare with a theoretical density:

```
histogram(x, 35, 'Normalization', 'pdf')
xlabel('Value')
ylabel('Estimated probability density')
grid on
```

- estimate any probability by counting the samples satisfying the condition and dividing by `numel()`:

```
p_estimated = sum(x >= a & x <= b)/numel(x);
```

2.3 Summary: functions and notation you should know

Signature or notation	Example	Description
$A \sim \mathcal{N}(\mu, \sigma^2)$	$A \sim \mathcal{N}(2, 2)$	Normal distribution: mean μ , variance σ^2
$\sigma = \sqrt{\sigma^2}$	$\sigma = \sqrt{2}$	Standard deviation, used to scale <code>randn()</code>
<code>randn(m,n)</code>	<code>randn(1,1000)</code>	$m \times n$ samples from $\mathcal{N}(0, 1)$
<code>mu + sigma*randn(m,n)</code>	<code>2 + sqrt(2)*randn(1,1000)</code>	Samples from $\mathcal{N}(\mu, \sigma^2)$
<code>mean(x), std(x), var(x)</code>	<code>std(x)</code>	Sample mean, standard deviation and variance
<code>histogram(x,n)</code>	<code>histogram(x,35)</code>	Histogram with n bins; add 'Normalization', 'pdf' for unit area
<code>erf(x)</code>	<code>0.5*(1+erf((x-mu)/(sigma*sqrt(2))))</code>	Error function used to evaluate the normal CDF
$F_A(b) - F_A(a)$	$P(a \leq A \leq b)$	Interval probability from the CDF
<code>sum(condition)/numel(x)</code>	<code>sum(x > 5)/numel(x)</code>	Empirical probability

3 Theoretical exercises

1. Let A be a continuous random variable with the normal distribution $\mathcal{N}(\mu = 1, \sigma^2 = 20)$.
 - a. Calculate the probability $P(A \in [2, 4])$.
 - b. What is the distribution of the random variable B defined as $B = A - 2$?
 - c. What is the maximum value of the distribution $w_A(x)$, and for what value x is it attained?
 - d. What is the distribution of the random variable C defined as $C = 3 * A$?(Hint: Start from $\mu_C = \overline{C}$ and $\sigma_C^2 = \overline{(C - \mu_C)^2}$, and replace C .)

4 Exercises

Write all commands yourself in a script or live script. Label figures and answer interpretation questions in comments.

1. Normal random samples:
 - a. Generate 10000 samples from $\mathcal{N}(\mu = 2, \sigma^2 = 2)$ and store them in `x_normal_1`. The second parameter is the variance.
 - b. Plot the first 100 values against sample index.
 - c. Plot a 35-bin PDF-normalized histogram.
 - d. Compute the sample mean, standard deviation, and variance. Compare them with 2, $\sqrt{2}$, and 2.
 - e. Generate the vector again. Which results change?
2. The effect of the normal-distribution parameters:
 - a. Generate 10000 samples from $\mathcal{N}(\mu = -3, \sigma^2 = 2)$. Compare its PDF-normalized histogram with the one from Exercise 1.
 - b. Keep $\mu = 2$, but set $\sigma = 0.5$. Generate 10000 samples and compare again.
 - c. Describe the changes in location, width, and height in each comparison.
 - d. Compute the sample mean and standard deviation for the two new vectors. Compare them with their theoretical values.

3. Probabilities for a normal variable:

Let $A \sim \mathcal{N}(\mu = 2, \sigma^2 = 2)$.

- a. Use `erf()` to compute $P(A \leq 3)$, $P(0 \leq A \leq 4)$, and $P(A > 5)$.
- b. Estimate all three probabilities from `x_normal_1`.

- c. Repeat the estimates with 100000 samples. Compare with theory.
 - d. Compute the expected number of samples greater than 5 for $N = 10000$. Compare it with the observed count in `x_normal_1`.
4. Same mean and variance, different distributions:
- a. Generate 10000 uniform samples from $[2 - \sqrt{6}, 2 + \sqrt{6}]$.
 - b. Use the uniform-distribution formulas from Lab 2 to verify that its theoretical mean and variance are both 2.
 - c. Compare its PDF-normalized histogram with that of `x_normal_1`.
 - d. Compare their sample means and variances. Can these two quantities alone distinguish the distributions?
 - e. Estimate $P(1 \leq A \leq 3)$ for each vector. Are the probabilities similar?

4.1 Supplementary exercise

5. Estimate a PDF directly from samples:

Write `p = myPDF(v, x, epsilon)` using

$$\hat{w}_A(x) = \frac{\text{number of samples in } [x - \varepsilon, x + \varepsilon]}{2\varepsilon N}, \quad N = \text{numel}(v).$$

- a. Evaluate it at 100 points from -4 to 8 using `x_normal_1`.
- b. Plot the estimate together with the theoretical normal PDF. Use element-wise operations when evaluating the PDF on a vector.
- c. Repeat with $\varepsilon = 0.1, 0.3$, and 0.8 . Compare smoothness and detail.

5 Final questions

- 1. A generated normal vector has minimum and maximum values. Why should you not use them as fixed limits for future samples from the same distribution?
- 2. Could an experiment contain no samples above a threshold even when its theoretical exceedance probability is positive?
- 3. Why can two distributions with the same mean and variance give different probabilities for the same interval?
- 4. A histogram looks bell-shaped. Is that enough to conclude that the data follow a normal distribution? What additional comparisons could you make?